CHAPTER 11 · REFERENCE DEPTH

Radio Equipment & Operation

Everything so far has been theory; this chapter is your hands on the equipment. It covers the modes you communicate in, the discipline of the push-to-talk button, the controls on the panel, the antennas on the airframe, the emergency beacon that calls for help when you cannot, and the tell-tale signs that a radio has failed.

SYLLABUS MAP

Part II (iii)(iv) Communication systems · mic/squelch/AVC/tuning · simplex/duplex · advantages & limitations of RT

◎ Learning objectives — by the end of this chapter you will be able to…

- Distinguish simplex, duplex, half-duplex and broadcast, and state which aviation VHF uses.
- Apply correct push-to-talk technique and explain the danger of a stuck microphone.
- Describe microphones, headsets, ANR, the audio panel and intercom.
- Explain squelch, AVC/AGC, volume, gain and frequency selection.

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- Describe aircraft antennas, the need for matching, and the ELT and ACARS.
- Troubleshoot a radio failure and state the advantages and limitations of RT.

11.1 Simplex, duplex & half-duplex

11.2 Push-to-talk technique

11.3 Microphones, headsets & ANR

11.4 The audio panel & intercom

11.5 Squelch, AVC, volume & tuning

11.6 Antennas & matching

11.7 ELT & ACARS

11.8 Radio failure & troubleshooting

11.9 Advantages & limitations of RT

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Understanding the controls on the COM panel and the audio selector is essential for reliable communication.

11.1 Simplex, duplex & half-duplex

FIRST PRINCIPLES — ONE FREQUENCY OR TWO?

The number of frequencies a link uses, and whether both ends can talk at once, defines the mode. Aviation VHF deliberately uses the simplest and most disciplined arrangement so that everyone on a shared channel hears everyone else.

Communication Modes

Mode	Frequencies	How it works	Example
Simplex	**One**	**Transmit OR receive, one at a time (PTT); all stations share the channel**	**VHF ATC voice**
Duplex	Two	Transmit AND receive simultaneously (different frequencies)	Telephone; some satcom
Half-duplex	Two	Two frequencies but still one-at-a-time at the user end	Some HF / relay links
Broadcast (Simplex)	One	One station transmits, others only listen	ATIS, VOLMET

THE AVIATION FACT

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VHF aeronautical voice is simplex — a single frequency shared by all, with only one station transmitting at a time. **ATIS** is simplex broadcast (the ground transmits, aircraft only listen). This shared discipline is why listening-out and brevity matter so much.

11.2 Push-to-talk technique

✓ THE CORRECT SEQUENCE

1. **Listen out** first — make sure the frequency is clear.
2. **Press** the PTT and pause a beat so the first word isn't clipped.
3. **Speak** at a steady pace, normal volume, microphone close to the lips.
4. **Release** the PTT fully when finished so you can hear the reply.

⚠ EXAM TRAP — THE STUCK MICROPHONE

On simplex, **while you transmit you cannot hear**. A PTT held down or stuck (a "stuck mic") therefore **blocks the whole frequency for everyone** and the offender doesn't even know — nobody else can get a word in. Never hold the button needlessly, release it fully, and if a frequency goes silent suspect a stuck mic.

WORKED EXAMPLE — TWO AIRCRAFT TRANSMIT TOGETHER

Two aircraft press PTT at the same instant on the same simplex VHF frequency.

Because the channel carries **AM** (Chapter 10), the two carriers beat together and the

controller hears a **heterodyne squeal** rather than either message. The controller says "two together, say again", and each aircraft re-transmits.

With FM's capture effect one would have been silently lost — which is exactly why aviation uses AM.

11.3 Microphones, headsets & ANR

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WHAT EACH DOES

A **microphone** converts your voice to an electrical signal; **noise-cancelling** (differential) types reject the steady cockpit roar so only your speech is transmitted. A **boom or oxygen-mask microphone** keeps it positioned consistently. The **headset** reproduces received audio; **ANR (Active Noise Reduction)** headsets generate an anti-phase signal to cancel low-frequency cabin noise, reducing fatigue and improving intelligibility.

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COCKPIT REALITY

Hold the mic **close and steady** and speak across it, not into it, to avoid "popping". A consistent mic position is half of good readability — the other half is a steady, unhurried pace.

11.4 The audio panel & intercom

SELECTING WHO HEARS WHOM

The **audio control panel** selects which transmitter the PTT keys (the **transmit selector**) and which receivers you listen to (the **receive selectors** — COM1, COM2, NAV idents, etc.). The **intercom (interphone)** lets the crew talk to each other — and to cabin/PA — **without transmitting** on the radio.

Selecting the wrong transmitter is a common cause of "transmitting on the wrong box".

EXAM TRAP

An intercom conversation is **internal** — it does not go out on the radio. Conversely, keying the PTT transmits on whichever COM the **transmit selector** is set to, regardless of which receivers you are monitoring.

11.5 Squelch, AVC, volume & tuning

THE PANEL CONTROLS

Squelch — mutes the constant background hiss when no signal is present, opening the audio only when a real signal arrives. Set it just past the point where the hiss

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disappears.

AVC / AGC (Automatic Volume / Gain Control) — automatically holds the received audio at a steady level despite changing signal strength, so a distant station and a near one sound similar.

Volume — sets the listening level.

Tuning / frequency select — picks the operating channel, usually with a **standby/active** pair so the next frequency can be set ready and flip-flopped in.

⚠ EXAM TRAP — SQUELCH VS AVC

Set squelch **too tight** and weak (distant) signals are muted and missed. **AVC does the opposite job to squelch** — it keeps a wanted signal's *level* constant; it does not mute noise. Don't confuse the two.

11.6 Antennas & matching

MATCHING THE ANTENNA TO THE WAVE

The **VHF COM antenna** is the vertical blade — sized to a fraction of the wavelength and matched to the vertically-polarized VHF signal. **HF** uses a long wire or a probe (often in the fin) with an **antenna coupler/tuner**, because its long wavelength makes a full-size aerial impossible. A well-matched antenna transfers nearly all the transmitter power to the air; a mismatch reflects power back (high **VSWR**), wasting it and risking the transmitter.

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WORKED EXAMPLE — WHY THE VHF AERIAL IS A SMALL BLADE

At 120 MHz the wavelength is 2.5 m (Chapter 7). A common efficient length is a **quarter-wave:** ¼ × 2.5 m = **0.625 m** — a practical blade.

An HF signal at 3 MHz has λ = 100 m; a quarter-wave would be 25 m — impossible on an airframe, so HF uses a long-wire/probe and a coupler to make a short aerial behave as if resonant.

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COCKPIT REALITY — ANTENNA SHADOWING

In a steep turn the airframe can shadow an antenna mounted on the opposite side, briefly weakening the signal. It is one reason aircraft carry **two** COM antennas/radios and why a momentary drop in a hard turn is normal, not a fault.

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The short vertical blades on the fuselage are the VHF COM antennas, sized to match a quarter-wavelength of the signal.

11.7 ELT & ACARS

ELT — EMERGENCY LOCATOR TRANSMITTER

A beacon that activates **automatically on heavy impact** (g-switch) or manually, transmitting on **406 MHz** — detected by the Cospas-Sarsat satellite system, carrying a coded, **registered identity** so the aircraft is known — and on **121.5 MHz** for short-range homing by rescuers. It is carried under Annex 6 (Chapter 3). Functional testing is restricted to defined windows to avoid false alerts. (Full detail in Chapter 22.)

ACARS & DATALINK

ACARS (Aircraft Communications Addressing and Reporting System) is a digital datalink for short **text** messages between aircraft and ground — clearances, position reports, weather, maintenance and operational messages — reducing voice workload. **CPDLC** (Controller–Pilot Data Link Communications) carries ATC clearances as text, used especially in oceanic airspace (Chapter 18).

11.8 Radio failure & troubleshooting

QUICK CHECKS BEFORE DECLARING FAILURE

Hearing nothing / getting no replies? Work through: correct **frequency** selected (active, not standby)? · **volume** up and **squelch** not too tight? · correct **transmit selector** and **receive selectors** on the audio panel? · **headset/PTT** plugged in and the PTT not

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stuck? · **circuit breaker / power** in? · try the **second COM**. If still nothing, follow the communications-failure procedure (Chapter 20) and **squawk 7600**.

MNEMONIC

"Frequency, Volume, Squelch, Selector, Plug, Power" — the six things to check before believing the radio has truly failed. Then squawk **7600**.

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11.9 Advantages & limitations of RT

Advantages and Limitations of VHF RT

Advantages	Limitations
Instant, direct two-way voice	VHF range limited to line-of-sight
Simple, universal; no fixed infrastructure airborne	One-at-a-time on simplex — congestion on busy frequencies

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Advantages	Limitations
Standard phraseology crosses language barriers	Subject to interference, fading and ground shadow
Immediate read-back confirms understanding	Voice can be misheard — discipline essential
Datalink (ACARS/CPDLC) offloads routine traffic	A stuck mic can block a whole channel

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☆ Numbers to memorise

Essential Facts for Chapter 11

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Fact	Value
Aviation VHF voice	Simplex — one frequency, one transmitter at a time (PTT)
ATIS / VOLMET	Simplex broadcast (listen only)
Squelch	Mutes background hiss until a real signal arrives

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Fact	Value
AVC / AGC	Keeps received audio level steady (opposite job to squelch)
VHF antenna	Vertical blade, ~quarter-wave; HF uses long-wire + coupler
ELT	406 MHz (satellite, registered ID) + 121.5 MHz (homing)
Radio-failure squawk	7600

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Question bank

Part A — MCQs (click an option to check)

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1. Aviation VHF voice communication operates in which mode?

Simplex

Duplex

Full-duplex

Broadcast only

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Answer: Simplex.

A single shared frequency, one transmitter at a time — simplex (PTT).

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2. ATIS is an example of:

Duplex

Simplex broadcast (listen only)

Half-duplex

Two-way simplex

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Answer: Simplex broadcast (listen only).

ATIS is a one-way broadcast; aircraft only listen.

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3. Duplex operation uses:

One frequency, one at a time

Two frequencies, allowing simultaneous transmit and receive

No frequency

Only HF

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Answer: Two frequencies, allowing simultaneous transmit and receive.

Duplex uses two frequencies so both ends can talk at once.

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4. A "stuck microphone" on a simplex frequency:

Improves the signal

Blocks the whole frequency for everyone

Only affects that aircraft

Changes the frequency

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Answer: Blocks the whole frequency for everyone.

A held/stuck PTT transmits continuously, blocking the shared channel; the offender can't hear it.

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5. The correct PTT sequence is:

Speak, press, release, listen

Listen, press & pause, speak, release fully

Press and hold throughout the exchange

Press and speak instantly

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Answer: Listen, press & pause, speak, release fully.

Listen out, press and pause, speak clearly, then release fully to hear the reply.

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6. A noise-cancelling microphone:

Increases transmitter power

Rejects steady cockpit noise so mainly speech is transmitted

Tunes the radio

Replaces the antenna

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Answer: Rejects steady cockpit noise so mainly speech is transmitted.

It cancels the steady ambient roar, transmitting clearer speech.

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7. The intercom (interphone) is used to:

Transmit to ATC

Let the crew talk to each other without transmitting on the radio

Broadcast ATIS

Tune the navigation receiver

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Answer: Let the crew talk to each other without transmitting on the radio.

Intercom is internal; it does not go out on the radio.

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8. The function of squelch is to:

Increase transmitter power

Mute background noise until a signal is received

Change frequency automatically

Keep audio level constant

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Answer: Mute background noise until a signal is received.

Squelch silences the hiss when no signal is present; keeping level constant is AVC's job.

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9. AVC (automatic volume control) acts to:

Mute weak signals

Hold the received audio at a steady level despite signal variation

Boost transmit power

Select the frequency

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Answer: Hold the received audio at a steady level despite signal variation.

AVC/AGC keeps the wanted signal's level steady as strength changes.

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10. Setting the squelch too tight will:

Increase range

Mute weak (distant) signals so they are missed

Boost the audio

Have no effect

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Answer: Mute weak (distant) signals so they are missed.

Too-tight squelch keeps the audio closed to weak but wanted signals.

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11. The VHF communication antenna is mounted:

Horizontally

Vertically, to match the vertically-polarized VHF signal

Inside the cabin

At 45°

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Answer: Vertically, to match the vertically-polarized VHF signal.

Aviation VHF is vertically polarized, so the blade is vertical.

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12. A poorly matched antenna causes:

More radiated power

Power reflected back (high VSWR), wasted and risking the transmitter

A lower frequency

Better range

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Answer: Power reflected back (high VSWR), wasted and risking the transmitter.

A mismatch reflects power (high VSWR) instead of radiating it.

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13. HF aerials need a coupler/tuner because:

HF is line-of-sight

HF wavelengths are too long for a full-size aerial on an airframe

HF uses AM

HF needs no antenna

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Answer: HF wavelengths are too long for a full-size aerial on an airframe.

A 3 MHz quarter-wave is 25 m; a coupler makes a short aerial behave as resonant.

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14. An ELT transmits on:

118.0 MHz only

406 MHz (satellite) and 121.5 MHz (homing)

1030 MHz

75 MHz

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Answer: 406 MHz (satellite) and 121.5 MHz (homing).

406 MHz for satellite detection with coded ID, plus 121.5 MHz for homing.

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15. ACARS is best described as:

A navigation aid

A digital datalink for short text messages between aircraft and ground

An emergency beacon

A type of antenna

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Answer: A digital datalink for short text messages between aircraft and ground.

ACARS carries short text messages, reducing voice workload.

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16. On total radio failure, the transponder code to select is:

7500

7600

7700

2000

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Answer: 7600.

7600 indicates radio/communications failure (7500 hijack, 7700 emergency).

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17. Before declaring a radio failure you should first check:

The weather

Frequency, volume, squelch, audio-panel selectors, PTT/plug and power

The transponder code

The fuel

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Answer: Frequency, volume, squelch, audio-panel selectors, PTT/plug and power.

Work the quick checks; many "failures" are a wrong frequency, tight squelch or wrong selector.

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18. A limitation of VHF radiotelephony is:

Unlimited range

Range restricted to line-of-sight

It needs no frequency

It cannot be misheard

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Answer: Range restricted to line-of-sight.

VHF is line-of-sight; range grows with altitude but is horizon-limited.

Part B — Oral / viva (tap to reveal model answers)

▼ ? **What is the difference between simplex and duplex, and which does aviation VHF use?**

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Model Answer:

Simplex uses one frequency with transmission and reception one at a time (PTT); duplex uses two frequencies allowing simultaneous transmit and receive. Aviation VHF voice is simplex; ATIS/VOLMET are simplex broadcasts.

▼ ? **Describe correct PTT technique and the hazard of a stuck microphone.**

Model Answer:

Listen out, press the PTT and pause briefly, speak clearly at a steady pace, then release fully.

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A stuck or held PTT transmits continuously, blocking the shared frequency for everyone — and the offender cannot hear it, so it must be guarded against.

▼ ? What do squelch and AVC do, and how do they differ?

Model Answer:

Squelch mutes the background hiss until a wanted signal is received; AVC (automatic volume/gain control) holds the received audio at a steady level despite changing signal strength. Squelch silences noise; AVC steadies a wanted signal — opposite jobs.

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▼ ? Why is the VHF antenna a short vertical blade while HF needs a coupler?

Model Answer:

Antennas are sized to a fraction of the wavelength. At VHF the wavelength is a few metres, so a quarter-wave blade is short and vertical to match the vertical polarization. HF wavelengths are tens to hundreds of metres, too long for a full aerial, so a long-wire/probe with a coupler is used to make a short aerial resonant.

▼ ? What is an ELT and on what frequencies does it transmit?

Model Answer:

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An Emergency Locator Transmitter, activating automatically on impact, transmitting on 406 MHz (satellite, with a coded registered identity) and 121.5 MHz (for homing) to aid search and rescue.

▼ ? State two advantages and two limitations of RT.

Model Answer:

Advantages: instant direct two-way voice; standard phraseology crossing language barriers; read-back confirms understanding. Limitations: VHF line-of-sight range; one-at-a-time

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congestion on simplex; susceptibility to interference, fading and ground shadow, and a stuck mic blocking the channel.

⚡ 60-SECOND REVISION CARD

Simplex (one freq, PTT) = aviation VHF; duplex = two freqs simultaneous; ATIS = simplex broadcast.
PTT: listen → press & pause → speak → release fully. A **stuck mic** blocks the channel.
Squelch mutes hiss; **AVC** steadies a wanted signal (opposite jobs). Intercom is internal.
VHF antenna = **vertical blade (~¼ wave)**; HF = long-wire + coupler; mismatch = high VSWR.
ELT 406 + 121.5; radio failure → check freq/vol/squelch/selector/plug/power → squawk **7600**.

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